

Yann Chaubet

Positions

- May 2026 **Visiting researcher**, Melbourne University.
2023– **Assistant Professor**, Nantes Université.
2022–2023 **Herchel Smith Research Fellow**, University of Cambridge.
By-Fellow of Churchill college

Education

- 2019–2022 **P.h.D in fundamental mathematics**, under the supervision of Colin Guillarmou, Université Paris-Saclay.
2014–2019 **École normale supérieure**, Paris.

Publications and Preprints

Minimax spectral estimation of weighted Laplace operators, with Vincent Divol.
preprint arXiv :2511.22694

Dynamical zeta functions, book chapter in *Microlocal analysis in hyperbolic dynamics and geometry* by Thibault Lefeuvre.
To appear in *Cours spécialisés de la SMF*.

Nonlinear chaotic Vlasov equations, with Daniel Han-Kwan and Gabriel Rivière.
preprint arXiv :2407.04426

Combinatorial zeta functions for triangles, with Léo Bénard, Nguyen Viet Dang and Thomas Schick.
preprint arXiv :2303.11226

Geodesic Lévy flights and expected stopping time for random searches, with Yannick Guedes Bonthonneau, Thibault Lefeuvre and Leo Tzou.
Probability theory and related fields, 2025, vol. 191, no 1, p. 235–285

Dynamical zeta functions for billiards, with Vesselin Petkov.
To appear in *Annales de l'Institut Fourier*.

Resolvent of vector fields and Lefschetz numbers, with Yannick Guedes Bonthonneau.
Pure and Applied Analysis, Vol. 6 (2024), No. 2, 521–540

Dynamical torsion for contact Anosov flows, with Nguyen Viet Dang.
Analysis & PDE, Vol. 17 (2024), No. 8, 2619–2681

Closed geodesics with prescribed intersection numbers.
Geometry & Topology, 28 :701–758, 2024

Closed billiard trajectories with prescribed bounces.
Annales Henri Poincaré, Vol. 24, No. 3, pp. 1051–1075, 2022

Poincaré series for surfaces with boundary.
Nonlinearity, 35(12) :5993, 2022

Awards and grants

- 2025 **International Emerging Action**, partnership with Melbourne University, 28 k€.
- 2025 **PEPS**, 4,4 k€.
- 2023 **PULSAR**, 10 k€.
- 2022 **Herschel Smith Research Fellowship**, 15 k€/year for three years.

Organization of conferences

- 2026 **Workshop : analyse semiclassique le long de la Loire**, Nantes.
- 2025 **Journées brestoises des jeunes chercheuses et chercheurs en analyse microlocale**, Brest.
- 2025 **Three days workshop on geometry, analysis and dynamics**, Longeville-sur-Mer.
- 2024 **ANR Adyct workshop**, Nantes.
- 2024 **Workshop on pseudo-holomorphic curves**, Teyran.

Invited talks

International conferences

- May 2026 **Three days workshop in Dynamics, Geometry and Analysis**, Melbourne.
- Déc 2025 **Conférence Géométrie et Dynamique**, Aussois. Mini-cours (3x1h).
- Sep 2024 **Rigidity phenomena in dynamics and spectral theory**, Zurich.
- Nov 2023 **Workshop : Hyperbolic dynamics and semiclassical analysis**, Jussieu.
- Jun 2023 **Analytic techniques in Dynamics and Geometry**, Les Diablerets.
- Mar 2023 **Rencontre aléatoire, dynamique et spectre**, Grenoble.
- Mar 2023 **École de printemps géométrie et dynamique**, Lille.
- Mar 2023 **Paris-London Analysis seminar**, London.
- Nov 2022 **Geometrical Inverse Problems**, Linz.
- Mar 2022 **Rigidity problems in geometry**, Roscoff.
- Feb 2022 **Spectra and Dynamics on (Locally) Symmetric Spaces**, Paderborn.
- Dec 2021 **Spectral Geometry in the clouds**, Online.
- Nov 2021 **ANR "Aléatoire, dynamique et spectre" meeting**, Nantes.
- Sep 2021 **Hyperbolic dynamical systems and resonances**, Porquerolles.
- Apr 2021 **Resonances in dynamics and semiclassical analysis**, Lausanne.
- Apr 2020 **Spectral problems, hyperbolic dynamical systems and quantum chaos**, Online.
- Oct 2018 **NxSE Tech Forum**, Paris.

Seminars

- May 2026 **Pure mathematics seminar**, University of Queensland.
- May 2026 **Analysis, PDE and Geometry seminar**, Monash University.
- Oct 2025 **Journée Rennes-Nantes d'analyse**, Rennes. Mini-cours (2x1h).
- Mar 2025 **Dynamical Systems Seminar**, Pise.
- Jan 2025 **Séminaire de dynamique, probabilités et statistique**, Brest.
- Nov 2024 **Séminaire de géométrie**, Tours.
- Fev 2024 **Séminaire Quimpériodique**, Quimper.
- Jan 2024 **Séminaire Topologie, Géométrie et Algèbre**, Nantes.
- Jun 2023 **Ergodic Theory and Dynamical Systems Seminar**, Bristol.
- Mar 2023 **Séminaire Géométrie et Analyse Globale**, Nantes.

Fev 2023 **London Analysis and Probability seminar**, London.
Fev 2024 **Ergodic Theory and Dynamical Systems Seminar**, Warwick.
Feb 2023 **Workgroup on rigidity problems**, Cambridge University.
Jan 2023 **Differential geometry and topology seminar**, Cambridge University.
Dec 2022 **Séminaire Géométrie et Topologie**, Institut de Mathématiques de Marseille.
Nov 2022 **Séminaire Géométrie et Topologie**, Institut de Mathématiques de Jussieu.
May 2022 **Séminaire Gaston Darboux**, Institut Montpellierain Alexander Grothendieck.
Apr 2022 **Oberseminar Dynamische Systeme**, Ruhr-Universität, Bochum.
Apr 2022 **Seminari di Sistemi Dinamici**, Centro di Ricerca Matematica Ennio De Giorgi, Pisa.
Feb 2022 **Séminaire de Physique Mathématique**, Institut de Mathématiques de Bordeaux.
Jun 2021 **Séminaire de géométrie**, Institut de Mathématiques d'Orsay.
Apr 2021 **Geometrische Analysis und Zahlentheorie**, Universität Paderborn.
Mar 2021 **Séminaire de vulgarisation des doctorants**, Institut de mathématiques d'Orsay.
Nov 2020 **Séminaire d'analyse des doctorants**, Institut de mathématiques d'Orsay.
Fev 2020 **Séminaire d'Analyse Harmonique**, Institut mathématique d'Orsay.
Jan 2020 **Séminaire Teich**, Institut mathématique de Marseille.
Fev 2019 **Working group : b-calculus**, Institut mathématique d'Orsay.